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# UNIVERSITY OF KARACHI

## Departmental Test for Admission in B.S. (Computer Science)

Each question is followed by the five choices. Choose the best and write corresponding letter (A - E) on the ANSWER SHEET.

- Q.01. The amount of radioactive material remaining after a period of time is related to the  
(A) atomic mass (B) pressure (C) half-life  
(D) volume (E) none of these
- Q.02. A virtual image may be :  
(A) larger than the object (B) smaller than the object (C) erect  
(D) all of the above (E) none of the above
- Q.03. An object 3.00 cm tall is placed 24.0 cm from a converging lens. A real image is formed 8.00 cm from the lens. The focal length of the lens is :  
(A) 4.0 cm (B) 5.0 cm (C) 5.5 cm  
(D) 8.0 cm (E) 6.0 cm
- Q.04. The force necessary to produce an acceleration of  $6.00 \text{ m/s}^2$  on a mass of 5.00 Kg is :  
(A) 30.0 N (B) 31.0 N (C) 32.0 N  
(D) 33.0 N (E) 35.0 N
- Q.05. Accuracy is:  
(A) The same as precision (B) The number of significant digits  
(C) The smallest unit with which a measurement is made (D) All of the above  
(E) None of the above
- Q.06. A 60.0g bullet is fired from a gun and has 3150J of kinetic energy. Its velocity is  
(A) 325 m/s (B) 324 m/s (C) 300 m/s  
(D) 260 m/s (E) 400 m/s
- Q.07. A force of 5.0 N is applied to a spring whose elastic constant is 0.250 N/cm. Its change in length is  
(A) 20 Cm (B) 30 Cm (C) 40 Cm  
(D) 25 Cm (E) 23 Cm
- Q.08. The amount of heat required to melt 1 Kg of a solid is called its  
(A) heat of vaporization (B) mass density (C) weight density  
(D) heat of fusion (E) latent heat
- Q.09. Two charges, each with magnitude  $+6.50 \times 10^{-5} \text{ C}$ , are separated by a distance of 0.200 cm. The force of repulsion between them is  
(A)  $9.51 \times 10^6 \text{ N}$  (B)  $9.34 \times 10^6 \text{ N}$  (C)  $9.44 \times 10^6 \text{ N}$   
(D)  $8.56 \times 10^6 \text{ N}$  (E)  $9.22 \times 10^6 \text{ N}$
- Q.10. The resistance of a copper wire 20.0 m long with cross-sectional area of  $6.56 \times 10^{-3} \text{ cm}^2$  at  $20^\circ \text{C}$  is (The resistivity of copper at  $20^\circ \text{C}$  is  $1.72 \times 10^{-6}$ )  
(A) 1 ohm (B) 0.5 ohm (C) 0.524 ohm  
(D) 1.5 ohm (E) 0.562 ohm
- Q.11. The current in a parallel circuit is given by  
(A) the sum of the inverse currents (B) the sum of the voltages  
(C) the sum of the currents in the branches (D) all of the above (E) none of the above

- Q.12. A hand drill draws 4.0 Ampere and has a resistance of 14.6 ohm. The power it uses is
- (A) 150 watts (B) 200 watts (C) 233 watts  
(D) 234 watts (E) 300 watts
- Q.13. The maximum voltage in an alternating current is 75 V. The instantaneous voltage at  $\theta=35^\circ$  is
- (A) 45V (B) 34V (C) 43V  
(D) 46V (E) 50V
- Q.14. The capacitive reactance of a 10.0  $\mu\text{F}$  capacitor in a circuit of frequency 1.00 KHz is
- (A) 15.9 ohm (B) 20 ohm (C) 15.5 ohm  
(D) 15.3 ohm (E) 16 ohm
- Q.15. The FM band of a radio is centered around a frequency of 100 Mhz. The wave length of radio transmission is
- (A) 2.5 m (B) 3.0 m (C) 4.0 m  
(D) 3.5 m (E) 4.5 m
- Q.16. If you are using MS Word and Netscape Navigator on your computer simultaneously this is called \_\_\_\_\_
- (A) Workgroup computing (B) Application integration (C) Processing handling  
(D) Multiprocessing (E) Multitasking
- Q.17. Which of the following is not a valid email address? {NOTE: xyz is someone's name, ku is his/her institution's name, des is his/her department's name and pk is his/her country's code}
- (A) xyz@ku.edu (B) xyz@ku.edu.pk (C) xyz@des.ku.edu  
(D) xyz@des.ku.edu.pk (E) xyz@des.ku
- Q.18. When an operating system uses secondary storage to store programs in RAM during execution of those programs, it is called \_\_\_\_\_
- (A) secondary memory (B) external memory (C) virtual memory  
(D) extended memory (E) expanded memory
- Q.19. An 8 bit computer word is to hold both negative and positive numbers with most significant bit representing a sign bit. The largest positive number that can be stored is
- (A) 1023 (B) 511 (C) 255  
(D) 127 (E) none of these
- Q.20. Bits are used to represent data inside a computer, these bits are stored in RAM as \_\_\_\_\_
- (A) zeros and ones (B) true and false values (C) logical values  
(D) low and high frequency signals (E) high and low voltage levels
- Q.21. A rule violation in a program is called \_\_\_\_\_ error.
- (A) semantic (B) language (C) programming  
(D) processing (E) syntax
- Q.22. Amongst the following \_\_\_\_\_ is used for converting printed documents into machine-readable form.
- (A) OMR (B) MICR (C) OCR  
(D) Barcode reader (E) PDF reader
- Q.23. Which of the following may not be a medium/technology for a wireless LAN?
- (A) radio waves (B) infrared (C) satellite  
(D) DSL (E) A and D are correct

Q.24. Which of the following is not a storage device ?

- (A) RAM (B) Cache (C) Flash card  
(D) PCB (E) RAM card

Q.25. Two binary numbers are to be subtracted. If A represents first number and B represents two's complement of the number which is to be subtracted then to subtract these numbers :

- (A) A and B are to be added  
(B) A and B are to be subtracted  
(C) two's complement of A is to be subtracted from B  
(D) two's complement of A is to be added into B  
(E) subtraction is not possible without converting B in its original form

Q.26. Which of the following are character-encoding schemes?

- 1) ASCII 2) ANSI 3) EBCDIC 4) Unicode 5) BCD

- (A) 1 and 3 only (B) All except 4 and 5 (C) All except 5  
(D) All (E) All except 2 and 5

Q.27. Which of the following is not a markup language?

- (A) HTML (B) XML (C) VRML  
(D) SGML (E) UML

Q.28. All server computers on the Internet have an associated domain name if a server does not have an associated domain name then \_\_\_\_\_.

- (A) can't be accessed by Internet users  
(B) can be accessed using their TCP/IP addresses  
(C) can only be accessed by directly dialing the modem of that server  
(D) can be accessed only through a satellite  
(E) can only be accessed by hackers.

Q.29. A three bit binary number is to be stored in excess-4 format. In this format numbers can be stored ranging from -4 to +3. To convert a number into excess-4 format, 4 is added to the number and result is converted into binary. So the bit pattern to store zero is \_\_\_\_\_.

- (A)  $000_2$  (B)  $001_2$  (C)  $100_2$  (D) other than A, B and C  
(E) Can't be converted because the number is neither positive nor negative

Q.30. The number  $110010_2$  is given in excess  $2^5$  form. What is the actual number?

- (A) 82 (B) 50 (C) 32  
(D) 18 (E) none of these

Q.31. WAP is acronym for \_\_\_\_\_.

- (A) Web Application Protocol (B) Web Access Point (C) Wireless Access Point  
(D) Wireless Application Protocol (E) None of these

Q.32. The product of  $01101110_2$  and  $0000011_2$  is

- (A)  $0000100100010010_2$  (B)  $0000001011011010_2$  (C)  $0000010100010110_2$   
(D)  $0000001100000010_2$  (E)  $0110111000000111_2$

Q.33. A software package needs a total of 6MB RAM. Base Memory of existing Hardware is 512 KB with no extended memory. Adding RAM chips will extend the memory to accommodate the software. If each RAM chip has a capacity of 256 KB find the number of chips to be used for extended memory?

- (A) Less than 21 (B) 21 (C) 22  
(D) 24 (E) none of these

Q.34. The number  $28_7 + 16_{16} =$  ?

(A)  $38_{10}$

(B)  $50_{10}$

(C)  $44_{10}$

(D) Number with different bases can't be added (E) can't be added since numbers are incorrect

Q.35. The truth table of Boolean expression  $(\bar{E} + C) \cdot \bar{D} \oplus (\bar{A} \cdot \bar{E})$  will contain \_\_\_\_\_ rows.

(A) 5

(B) 8

(C) 16

(D) 32

(E) none of these

Q.36. The graph of  $x = 5y^2 + 6$  is :

(A) Symmetric about x-axis

(B) Symmetric about the y-axis (C) symmetric about the origin

(D) not symmetric

(E) Symmetric about both axes simultaneously.

Q.37. Let a set be given as  $A = \{1, 2, 3\}$ . Some other sets are also given as follows:  $B = \{0, 1, 2, 3\}$ ,  $C = \{1, 2, 3, 4, \dots\}$ ,  $D = \{x \mid (x-3)(x^2-3x+2)=0\}$ ,  $E = \{3, 1, 2\}$ ,  $F = \{3, 2, 1\}$ . Then the following sets are equal to A

(A) D, E, F

(B) Only B

(C) Both B and D

(D) Both B and C

(E) Both B and F

Q.38. Let  $3 \leq 4 - 2x < 7$ , then

(A) x lies in the interval :  $-\frac{3}{2} \leq x < \frac{3}{2}$

(B) x lies in the interval :  $\left[-\frac{3}{2}, \frac{1}{4}\right]$

(C) x lies in the interval :  $-\frac{3}{2} < x \leq \frac{3}{2}$

(D) x lies in the interval :  $\left[-\infty, \frac{3}{2}\right]$

(E) x lies in the interval :  $\left[\frac{1}{2}, \infty\right]$

Q.39. Let  $y = 20 - 3u$  where  $u = 5x - 4$ , then the expression of  $\frac{dy}{dx}$  is equal to :

(A) 5

(B) -3

(C) -15

(D) 15

(E) -4

Q.40. Let  $f(x) = \frac{x^2 - 4}{x - 2}$ , then the limiting value of  $f(x)$  as x tends to 2 is equal to:

(A) 2

(B) 4

(C)  $\frac{0}{\infty}$

(D)  $\frac{0}{0}$

(E)  $\infty$

Q.41. The volume of a sphere, whose radius is given by "r" is :  $v = \frac{4}{3}\pi r^3$ . Let the relative error in radius be known as  $v = \frac{dr}{r} = \pm 0.0004$ . Then the relative error in volume which is  $\frac{dv}{v}$  is equal to

(A)  $\pm 0.016$

(B)  $\pm 0.00$

(C)  $\pm 0.0012$

(D)  $\pm 1.2$

(E)  $\pm 2.1$

Q.42. A function is given by the form :  $y = x^3 - 3x + 1$ . Then the values of x at which  $\frac{dy}{dx}$  is equal to zero are :

(A)  $\pm 1$

(B)  $\pm i$

(C)  $\pm 2i$

(D)  $\pm \sqrt{2}$

(E)  $\pm \sqrt{0.5}$

Q.43. Suppose  $y = \frac{1}{3}x^3 - \frac{3}{2}x^2 + 2x$ , then the slope of the tangent line at point (0, 1) is equal to :

(A)  $1/3$

(B) +2

(C) zero

(D) -2

(E)  $-1/3$

Q.44. The area under the curve  $y = 3 \cos 2x$  over the interval  $\left[0, \frac{\pi}{8}\right]$  is equal to:

- (A)  $\sqrt{2}$  (B)  $\frac{3\sqrt{2}}{4}$  (C)  $\frac{3}{3\sqrt{2}}$   
 (D)  $\frac{3}{2\sqrt{2}}$  (E)  $\frac{1}{\sqrt{2}}$

Q.45. Let  $\log_5(5^{2x}) = 8$ , then it can be solved for  $x$ , so that  $x$  is equal to

- (A) 4 (B) 25 (C)  $8(\log 5)^{-1}$   
 (D)  $1/8$  (E)  $1/2$

Q.46. The value of  $\frac{d}{dx}(\log_x e)$  is equal to:

- (A)  $\frac{+1}{(\log_e x)^2}$  (B)  $(\log_x e)^{-2}$  (C)  $(\log_e x)^2$   
 (D) 1 (E)  $\frac{-1}{x(\log_e x)^2}$

Q.47. The sum of the finite series:  $\sum_{k=1}^{50} \left( \frac{1}{k} - \frac{1}{k+1} \right)$  is equal to:

- (A) 1 (B)  $\frac{-1}{51}$  (C)  $\frac{50}{51}$   
 (D)  $\frac{1}{2}$  (E) 2

Q.48. A six-sided die is tossed giving outcomes, {1}, {2}, {3}, {4}, {5}, {6}. If the probability of each "odd outcome" is equal so that  $P(\{1\}) = P(\{3\}) = P(\{5\})$ ; and each even outcome has equal probability which is twice that of probability of odd outcome. Then the probability of each even outcome, is equal to:

- (A)  $1/3$  (B)  $1/9$  (C)  $2/9$   
 (D)  $2/3$  (E)  $3/18$

Q.49. Let  $A$  be the matrix:  $\begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}$ . Then  $A^2 - 2A + I$  is equal to

- (A)  $\begin{bmatrix} 1 & 0 \\ 4 & 0 \end{bmatrix}$  (B)  $\begin{bmatrix} 1 & 0 \\ 8 & 0 \end{bmatrix}$  (C)  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$   
 (D)  $\begin{bmatrix} -1 & 0 \\ 0 & 4 \end{bmatrix}$  (E)  $\begin{bmatrix} 0 & 1 \\ 0 & 4 \end{bmatrix}$

Q.50. Let a matrix be given as:  $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$  and an unknown matrix is:  $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$ . Then the matrix  $X$  when  $AX = I$  is equal to:

- (A)  $\begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix}$  (B)  $\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$  (C)  $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$   
 (D)  $\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$  (E)  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Q.51. If a coin is tossed twice and at the end of the second toss of the coin, a die is tossed, then the total number of outcomes in the sample space are :

- (A) 12 (B) 24 (C) 8  
(D) 36 (E) 72

Q.52. If A and B are two sets in a sample space S, then  $(A \cup B)'$  is equal to:

- (A)  $A' \cup B'$  (B)  $A \cup B$  (C)  $A \cap B$   
(D)  $A' \cap B'$  (E)  $A' \cap B'$

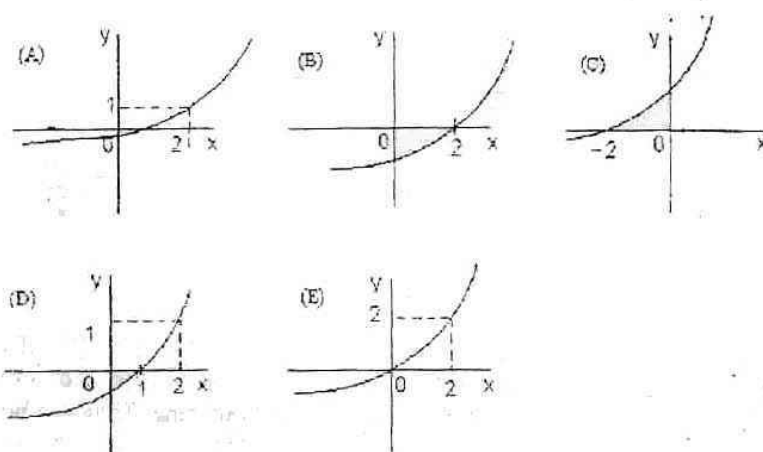
Q.53. Two points on a straight line are given: (24, 60000), (32, 44400). Then the equation of the line is:

- (A)  $y = 444x + 60000$  (B)  $y = 60x + 4444$  (C)  $y = 3x + 2$   
(D)  $y = 1950x + 13200$  (E)  $y = -1950x + 106800$

Q.54. Let  $f(x_1, x_2, x_3) = x_1^3 + 2x_1^2 x_2 - 3x_2 x_3 - 10$  then  $f(0, 2, -3)$  is equal to :

- (A) 8 (B) -28 (C) 18  
(D) 38 (E) 118

Q.55. If  $y = e^{x-2}$ , then the sketch of this function is:



Q.56. The solution of the system of equations:

$$\begin{aligned} 3x - 2y &= 6 \\ -15x + 10y &= -30 \end{aligned}$$

is given by :

- (A) infinitely many solutions (B)  $x = 1/3, y = 1/2$  (C)  $x = -1/15, y = -1/10$   
(D)  $x = -1/15, y = 1/10$  (E)  $x = 0, y = 0$

Q.57. The solution of the equation.  $\log_e x^2 + \log_e x = 9$  given by :

- (A)  $x = e$  (B)  $x = 27$  (C)  $x = 18$   
(D)  $x = e^3$  (E)  $x = 3$

Q.58. The value of  $\int_2^4 (4x - 5) dx + \int_2^4 (5 - 6x) dx$  is equal to:

- (A) +24 (B) -12 (C) 15  
(D) -10 (E) 16

Q.59. The function given by  $y = x^2 - 3x + 2$  is a:

- (A) straight line (B) hyperbola (C) parabola  
(D) ellipse (E) circle

Q.60. The maximum value of y in the equation  $y = 2x^2 + 4$  is attained at :

- (A)  $x = 2\sqrt{2}$  (B)  $x = \frac{1}{\sqrt{2}}$  (C)  $x = \sqrt{2}$   
(D)  $x = -4$  (E)  $x = 0$

- Q.61. Country code for dialing an international call to Pakistan is:
- (A) 21 (B) 42 (C) 91  
(D) 92 (E) 96
- Q.62. Pakistan's standard time is equivalent to:
- (A) GMT+5 (B) GMT-5 (C) GMT+4  
(D) GMT-4 (E) GMT+3
- Q.63. Which of the following countries is not included in SAARC:
- (A) India (B) Bangladesh (C) China  
(D) Sri Lanka (E) Nepal
- Q.64. Seoul is located in:
- (A) North Korea (B) South Korea (C) Malaysia  
(D) Indonesia (E) Japan
- Q.65. Which of the following country's currency is not called Dollar:
- (A) Australia (B) Hong Kong (C) Singapore  
(D) United States of America (E) Holland
- Q.66. Which of the following is not in England:
- (A) London (B) Surrey (C) Scotland  
(D) Liverpool (E) Exeter
- Q.67. Who was the first Pakistani Prime Minister to visit USA officially:
- (A) Liaquat Ali Khan (B) Zulfikar Ali Bhutto (C) Muhammad Khan Junejo  
(D) Benazir Bhutto (E) Nawaz Sharif
- Q.68. William Shakespeare died in the year:
- (A) 1515 (B) 1616 (C) 1717  
(D) 1818 (E) 1919
- Q.69. The famous English play "The Mousetrap" was written by :
- (A) Jules Verne (B) Charles Dickinson (C) Mark Twain  
(D) Agatha Christie (E) Enid Blyton
- Q.70. President of which country has been awarded the Nobel Peace Prize recently:
- (A) Japan (B) Iran (C) Singapore  
(D) France (E) South Korea
- Q.71. Which Pakistani cricketer got first married in the current year:
- (A) Saeed Anwar (B) Waqar Younus (C) Shahid Afridi  
(D) Saqlain Mushtaq (E) Yousuf Youhana
- Q.72. How many players participate in the match of basket ball from each side at a time:
- (A) 5 (B) 6 (C) 2  
(D) 10 (E) 11
- Q.73. The disease Rubella is also known as:
- (A) Typhoid (B) Influenza (C) Chicken pox  
(D) German measles (E) Malaria

Q.74. The largest clock of the world "Big Ben" is situated in which city of the world:

- (A) Paris (B) Rome (C) London  
(D) Moscow (E) New York

Q.75. Olympics 2000 were held in which country of the world:

- (A) England (B) USA (C) Canada  
(D) Japan (E) Australia

Fill the blanks with most appropriate words.

Q.76. If human beings will not provide them \_\_\_\_\_, they will find it for themselves.

- (A) their food (B) their clothes (C) with their food  
(D) their prey (E) none of these

Q.77. Richards was called \_\_\_\_\_ take someone else's place at the last minute.

- (A) on to (B) upon to (C) into  
(D) back to (E) to

Q.78. \_\_\_\_\_ there were only undeveloped stretches of coastline, hotels have \_\_\_\_\_, creating employment for local population.

- (A) where – built (B) where once – built (C) where once – sprung up  
(D) where – have built (E) where – have been sold

Q.79. Cautiously the men \_\_\_\_\_, from end to end, finding no one and detecting no signs of trouble.

- (A) searched it (B) traveled along (C) moved in  
(D) searched at (E) surveyed

Q.80. Although the critics agreed that the book was brilliant, so few copies were sold that the work brought the author \_\_\_\_\_.

- (A) little financial loss (B) great moral reward (C) small theoretical gain  
(D) little financial reward (E) great intellectual gain

Q.81. Opposite of the word scrappy is

- (A) sketching (B) bitty (C) disjointed  
(D) perfect (E) fragmentary

Q.82. He replied \_\_\_\_\_ ability.

- (A) far better than his (B) to the best of everyone's (C) to the best of his  
(D) in the best of his (E) with the best of his

Q.83. It's dull in our town \_\_\_\_\_ playmates left.

- (A) because (B) since my (C) till my  
(D) as soon as my (E) due to my

Q.84. This circumstance \_\_\_\_\_ the matters very serious.

- (A) certainly makes (B) certainly have made (C) certainly make  
(D) certainly has turned (E) have made

Q.85. Everything happened exactly \_\_\_\_\_ expected.

- (A) as was (B) like was (C) as had been  
(D) as have been (E) as has been

Q.86. In case \_\_\_\_\_, I shall email you.

- |                           |                      |                  |
|---------------------------|----------------------|------------------|
| (A) he was seriously sick | (B) he could be sick | (C) he fall sick |
| (D) he is seriously sick  | (E) he shall be sick |                  |

Q.87. He \_\_\_\_\_ the Taj Mahal when he went to Agra.

- |                |                    |               |
|----------------|--------------------|---------------|
| (A) may see    | (B) must see       | (C) might see |
| (D) must visit | (E) must have seen |               |

|                                                    |
|----------------------------------------------------|
| Choose answers that are closest to the given word. |
|----------------------------------------------------|

Q.88. Destitute

- |              |               |                 |
|--------------|---------------|-----------------|
| (A) wealthy  | (B) desperate | (C) destruction |
| (D) bankrupt | (E) destined  |                 |

Q.89. Ornate

- |                |              |             |
|----------------|--------------|-------------|
| (A) decorated  | (B) visible  | (C) evident |
| (D) attractive | (E) explicit |             |

Q.90. Lucid

- |            |           |             |
|------------|-----------|-------------|
| (A) remove | (B) clear | (C) explain |
| (D) liquid | (E) bold  |             |